Augmented Reality Guidance System: User Interface Flow Analysis

The system, labeled as **Munjaz AR Assistant**, provides guided assistance for troubleshooting and fixes, leveraging Augmented Reality to overlay digital instructions onto a physical environment. The flow is designed to ensure user safety and accuracy through distinct phases: Initialization, Environment Scanning, and Issue Detection.

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1. ⚙️ Session Initialization and Pre-Flight Checks

This phase prepares the user for the AR session and emphasizes critical operational and safety protocols.

Screen Name	Description	Key Elements & Function	Transition
Augmented Reality Guidance	This screen outlines the necessary prerequisites for a successful AR session, functioning as a system readiness and user safety checklist.	**Instructions include:** Holding the camera steady, pointing the camera at the target device/area, waiting for object detection, following visual markers, and moving slowly for correct device tracking. **Crucial Safety Note:** The session will automatically stop if the detected task is too complex or poses a danger, allowing the user to send a report or cancel.	Tapping **START SESSION** initiates the process. This action may trigger confirmation alerts (e.g., ensuring a clear camera view or warning about progress loss upon exit).
Alert (Pre-start)	A confirmation step that mandates a safety check before beginning the session.	It prompts the user to **Ensure your camera view is clear and the area is safe before starting.**	Tapping **Start** proceeds to the scanning phase.

2. 📡 Environment Scanning and Feature Detection

The system moves into the live camera view to establish the AR context and identify the area of concern.

Screen Name	Description	Key Elements & Function	Transition
Scanning Environment	The system actively engages the camera feed to analyze the environment.	The primary visual is an abstract, pulsating dot pattern representing the Spatial Mapping process (likely using SLAM - Simultaneous Localization and Mapping). **Instruction: Please keep your camera steady while the system analyzes the area.** This is essential for accurate world coordinate mapping.	Upon completion of the mapping and object identification, the system transitions to the issue detection outcome.
Water Leak Identified	A modal overlay indicating the successful detection and categorization of a specific problem.	**Message: water leak has been detected, follow the guidance to continue the safety check.** This confirms the successful detection of the *type* of issue and initiates a context-specific guidance sequence, starting with a safety verification.	The system proceeds to display AR overlays specific to the detected issue on the live camera feed.

3. 📍 Augmented Reality Issue Display

The final stage shows the AR system overlaying digital information onto the physical scene to guide the user.

Screen Name	Description	Key Elements & Function	Transition
Munjaz AR Assistant (Issue Detected)	This is the core AR guidance screen, displaying the real-time camera view of industrial pipes/machinery with overlaid digital markers.	**AR Markers:** Two distinct AR markers are visible, each pointing to a specific location on the physical structure. They display **Issue Detected** with a visual severity indicator (red/white blocks). This uses visual cues to draw the user's attention to the component(s) requiring intervention. **Action Buttons: Scan Again** (to restart the detection process) and	Tapping **Next** transitions the user to the first step of the guided repair sequence (not shown, but implied by the flow).

Screen Name	Description	Key Elements & Function	Transition
		Next (to proceed to the first guided repair step).	

This sequence of screens ensures that the user is adequately prepared, the environment is properly analyzed, and the specific issue is located and marked with clear digital guidance before any physical repair attempts are initiated.

9:41

Augmented Reality Guidance

This session uses augmented reality to guide you through simple IT fixes. Before you begin, please follow these instructions:

• Hold your camera steady.

• Po ant to tr

• Wa

• Fo at appear on your screen.

• Move slowly so the system can track your device correctly.

• If the system detects that the task is **too complex or dangers** , the session will stop and you can choose to send a report or cancel.

Alert

If you exit the session, all progress will be lost.

NO | Exit

START SESSION

9:41

Augmented Reality Guidance

This session uses augmented reality to guide you through simple IT fixes. Before you begin, please follow these instructions:

• Hold your camera steady.

• Point the camera at the device or area you want to troubleshoot.

• Wait for the system to detect the object.

• Follow the visual markers and instructions that appear on your screen.

• Move slowly so the system can track your device correctly.

• If the system detects that the task is **too complex or dangers** , the session will stop and you can choose to send a report or cancel.

START SESSION

9:41

Augmented Reality Guidance

This session uses augmented reality to guide you through simple IT fixes. Before you begin, please follow these instructions:

• Hold your camera steady.

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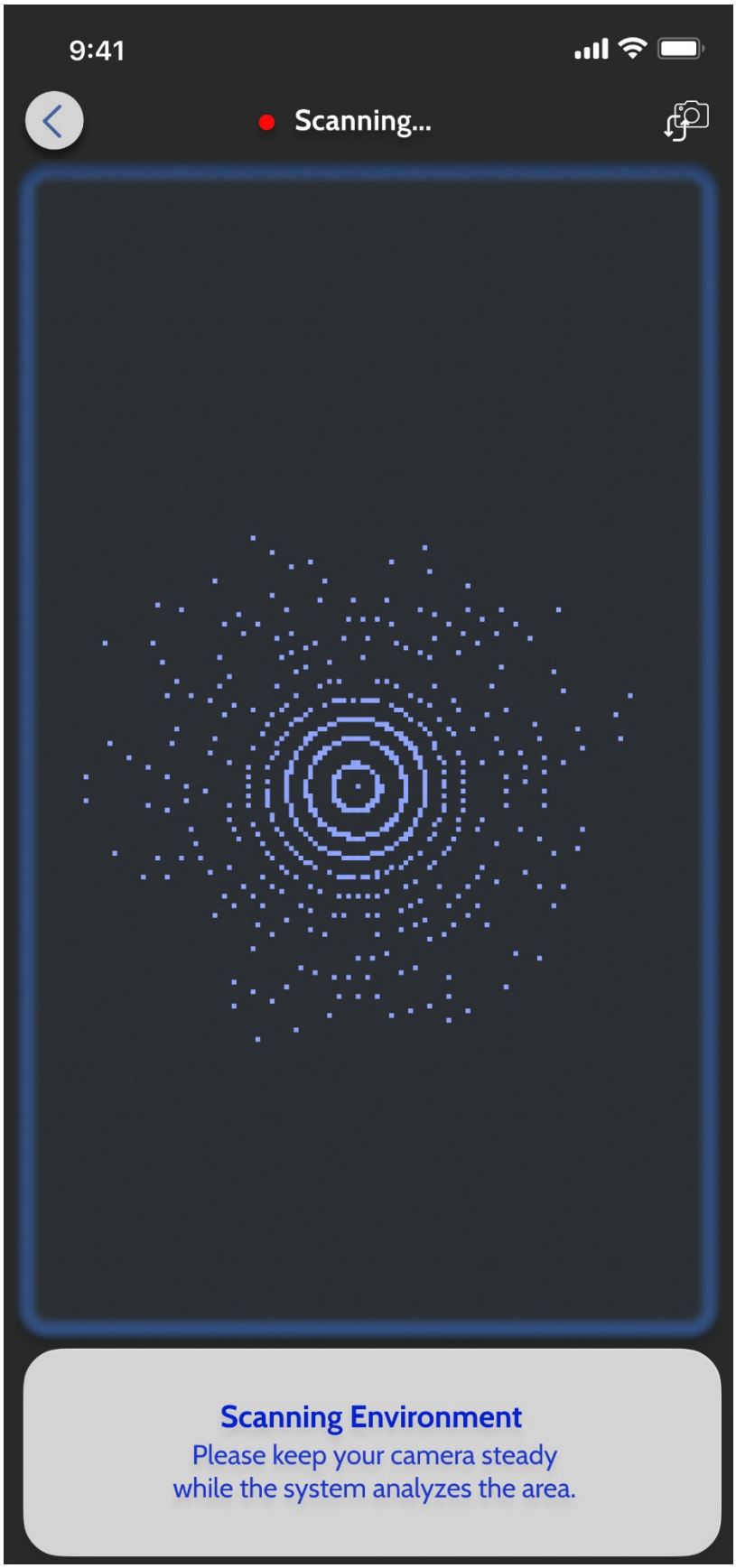
• If the system detects that the task is **too complex or dangers** , the session will stop and you can choose to send a report or cancel.

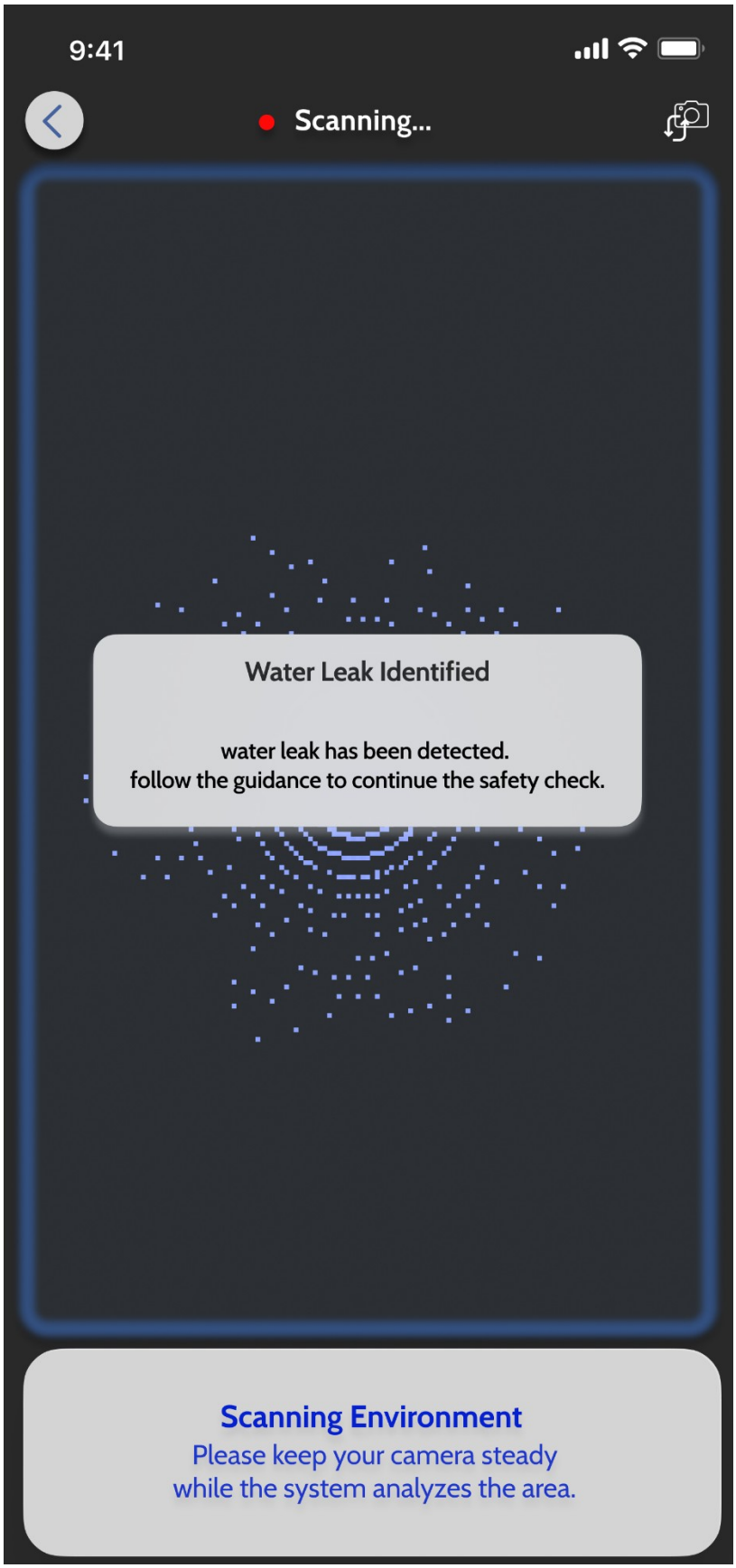
Alert

Ensure your camera view is clear and the area is safe before starting.

Start | Cancel

START SESSION


9:41
Scanning...
Scanning Environment
Please keep your camera steady
while the system analyzes the area.


9:41
Scanning...
Water Leak Identified
water leak has been detected.
follow the guidance to continue the safety check.
Scanning Environment
Please keep your camera steady
while the system analyzes the area.


9:41
Munjaz AR Assistant
Issue Detected
Issue Detected
Scan Again
Next

Path B: Safety and Escalation Protocol (Dangerous Case)

This sequence demonstrates the system's critical safety feature, where it overrides the repair guidance to prioritize user safety and trigger immediate escalation to professional support. This path is initiated based on the system's real-time environmental analysis, as warned in the initial session instructions.

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1. 🔍 Guided Safety Check (Pre-Danger Detection)

The system first attempts to verify the extent of the issue, even as it scans for surrounding hazards.

Screen Name	Description	Key Elements & Function	Transition
Munjaz AR Assistant (Step 1)	The system starts the guided investigation of the leak source.	**Guidance: Step 1: Move the camera closer to the floor to check the leak source.** This directs the user's focus, using the AR overlay to pinpoint the relevant area. The **Issue Detected** severity indicator remains visible.	Tapping **Next** transitions to the subsequent inspection step.
Munjaz AR Assistant (Step 2)	Continued guided inspection to scan components near the identified fault.	**Guidance: Step 2: Tilt the camera slightly left to inspect nearby components.** This confirms the system's attempt to gather comprehensive visual data before proceeding with the fix.	The data gathered during this inspection step triggers the safety override.

2. 🚨 Safety Override and Danger Alert

This is the system's immediate, critical response to the detection of a high-risk environmental hazard.

Screen Name	Description	Key Elements & Function	Transition
Danger Detected	The interface immediately shifts	**Warning Modal:** The modal text explicitly states the hazard	Acknowledging the danger

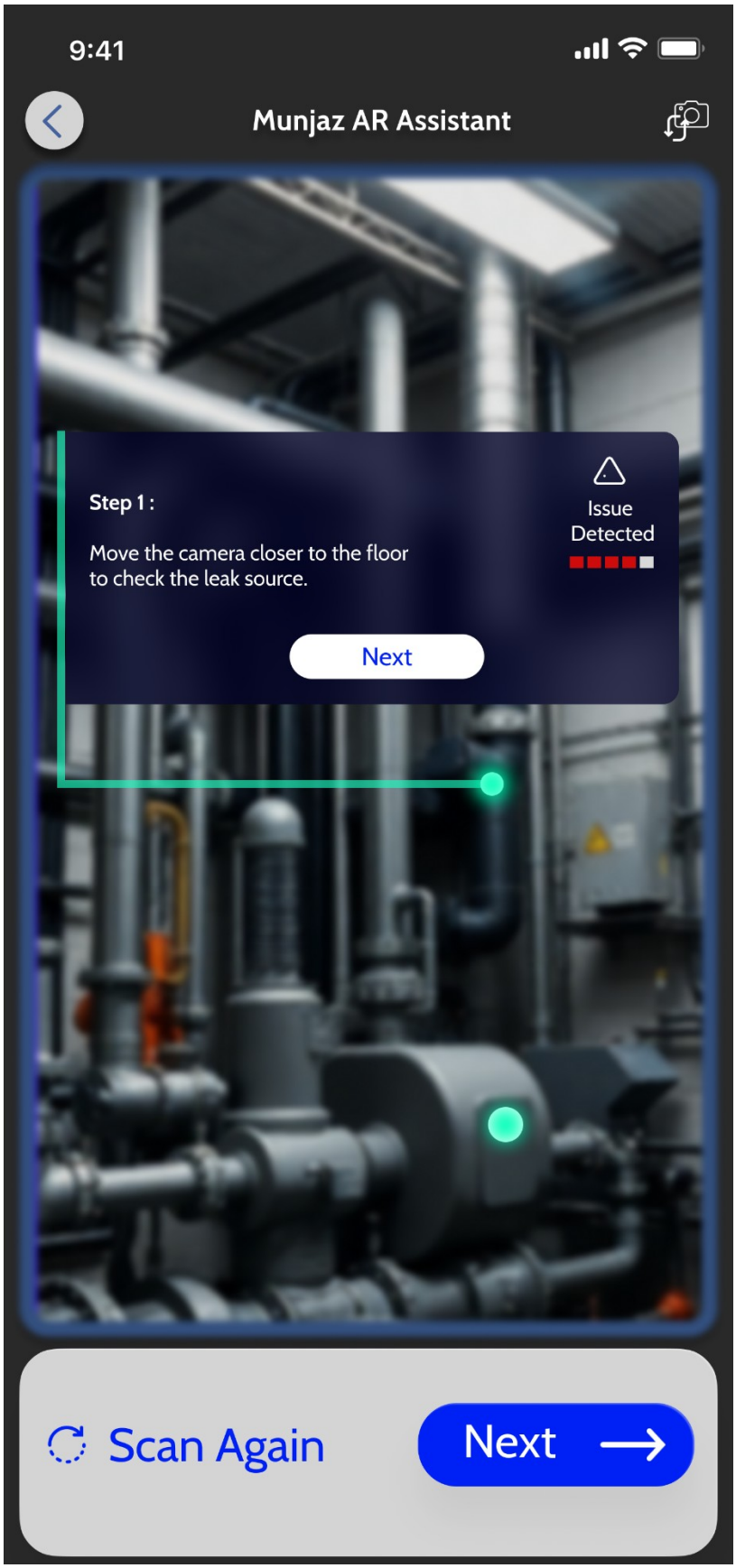
Screen Name	Description	Key Elements & Function	Transition
(Initial Warning)	to a **high-contrast red banner** labeled **Danger Detected** at the top. The AR guidance is stopped.	found: **This area contains exposed cables near the leak. It is unsafe to continue using AR guidance.** This demonstrates the system's ability to recognize complex, co-located environmental risks (water + electricity). The prompt is to tap **Next** (or **OK** in the red banner) to acknowledge the danger.	transitions to the final escalation and reporting stage.

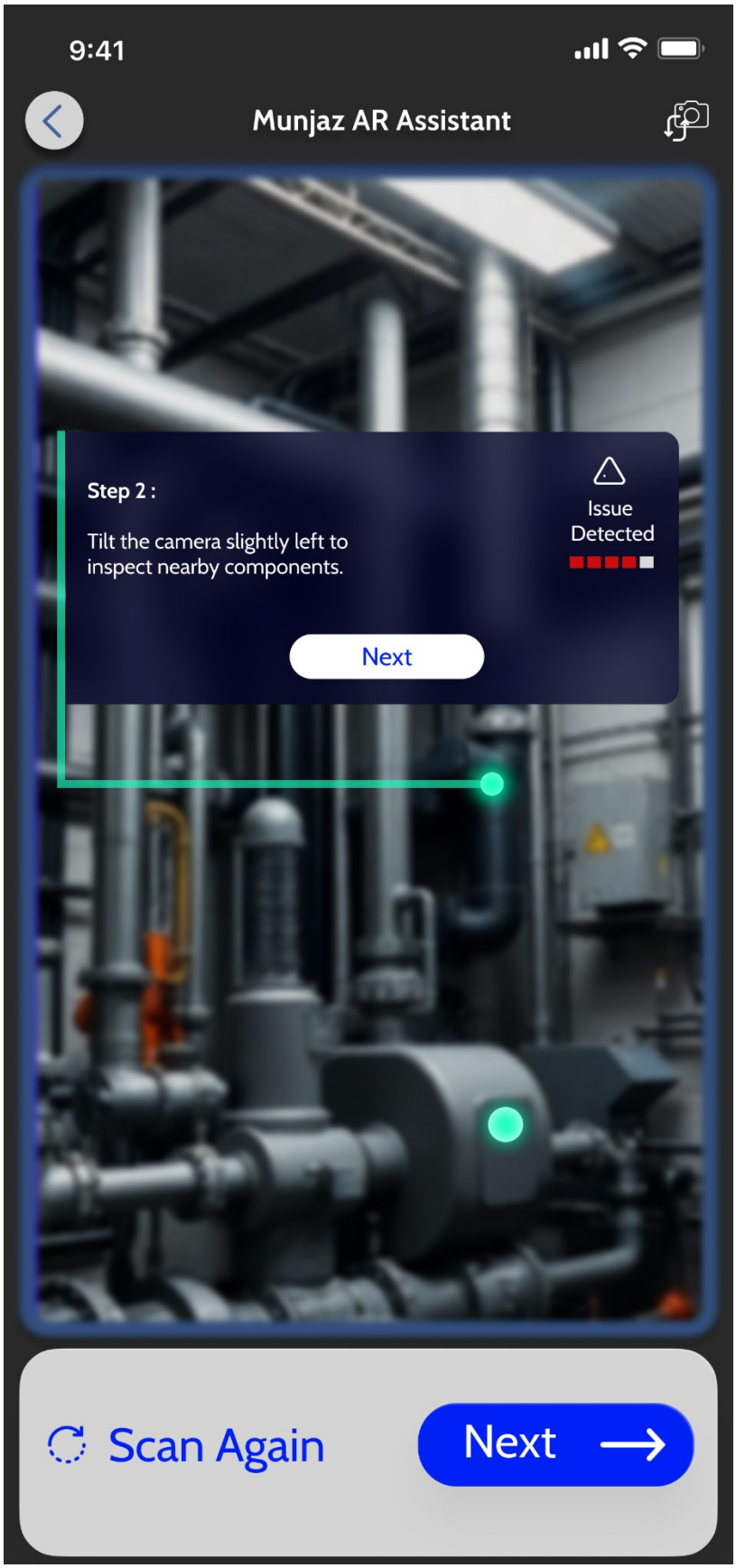
3. 📞 High-Risk Escalation and Expert Hand-off

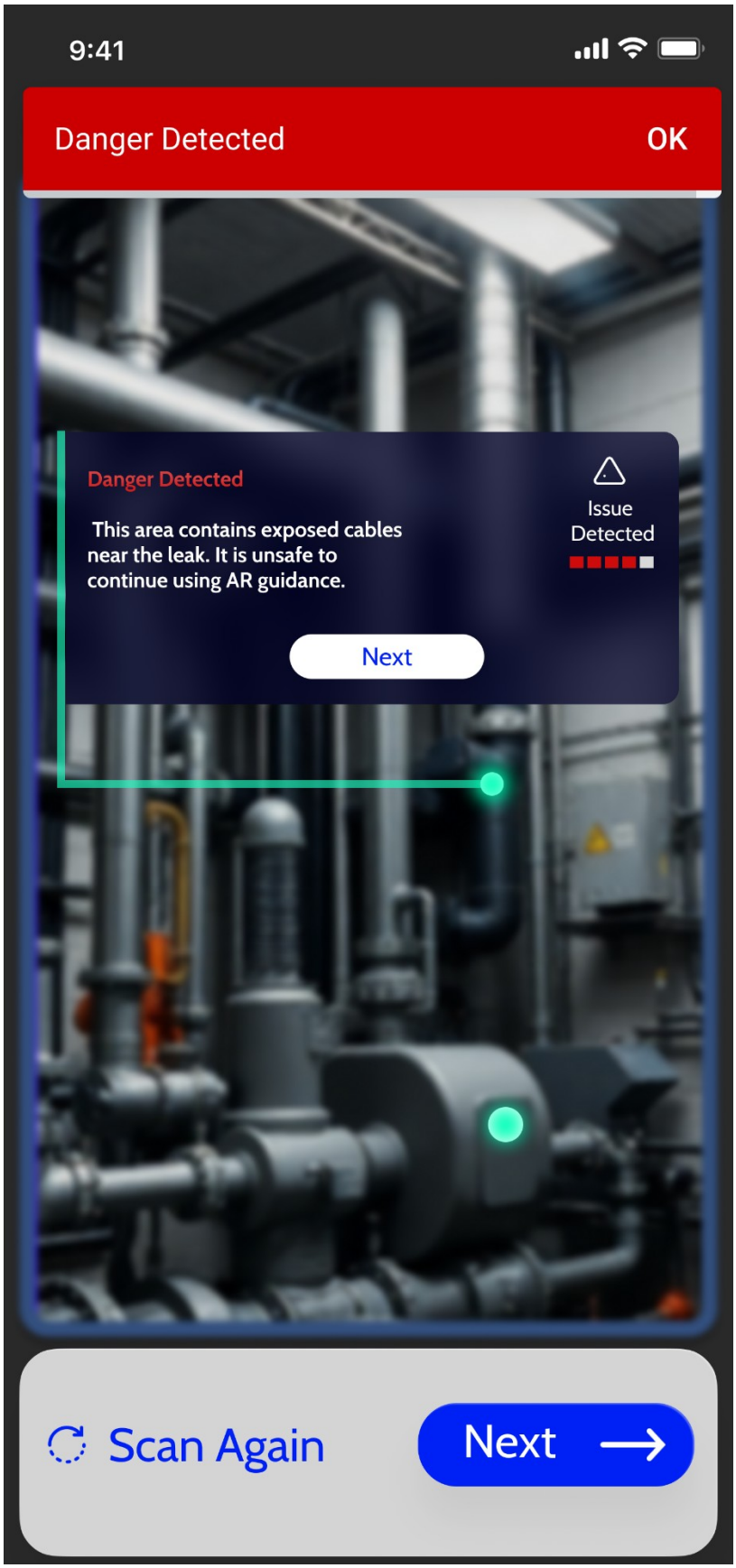
The final screen in this path ensures the user takes no further action and immediately contacts professional support.

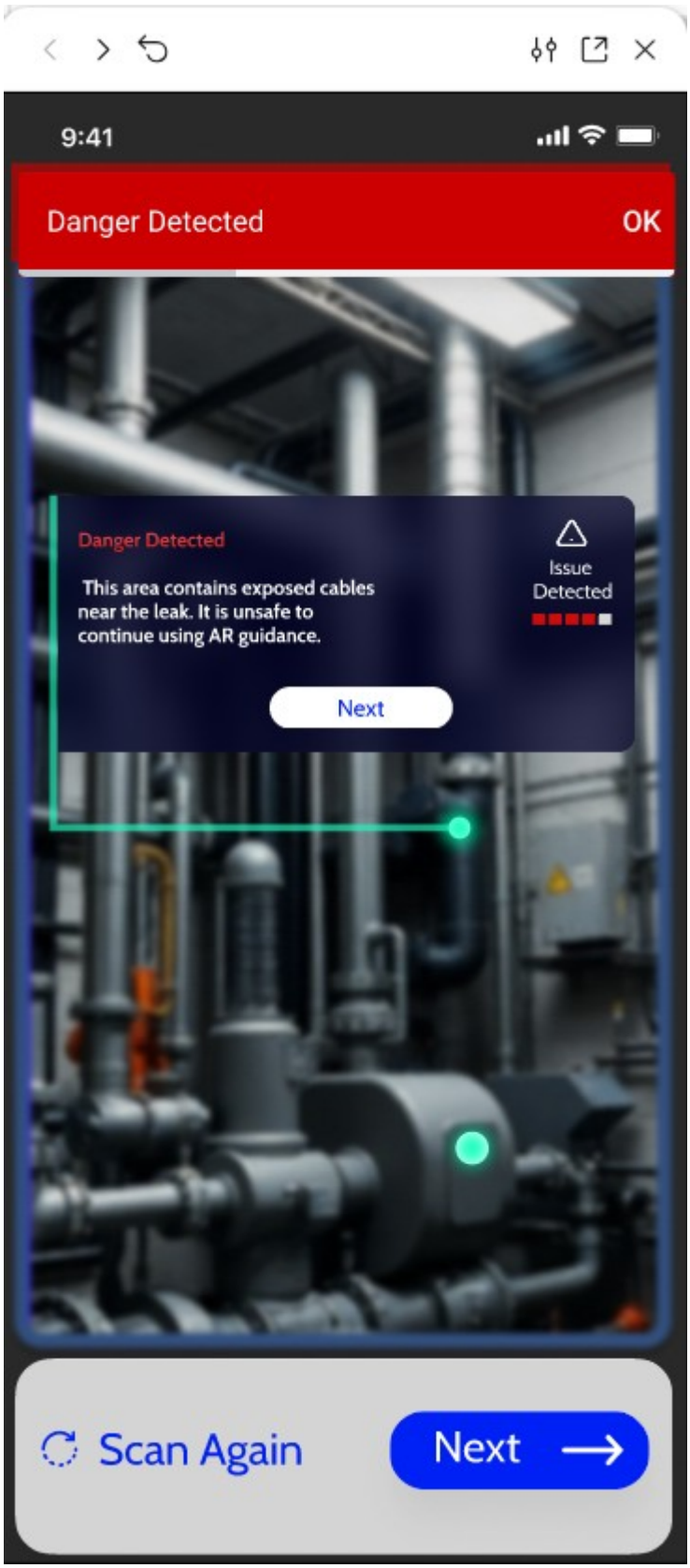
Screen Name	Description	Key Elements & Function	Outcome & Conclusion
Danger Detected (High Risk)	A final, definitive modal is presented over the scene, mandating cessation of activity and immediate contact with experts.	**Mandatory Action:** The **High Risk** warning instructs the user to **Stop immediately. Do not touch the area. Contact IT support. Escalation Button:** The **Send Report** button provides a mechanism to instantly transmit the system's gathered data (including images and the nature of the detected issue and danger) to the IT support team, ensuring that the arriving expert is fully informed.	The session is terminated, achieving the primary safety goal. The issue is officially escalated to the appropriate **IT support** personnel, ensuring the critical and dangerous issue is handled by an expert, as outlined in the initial safety guidelines.

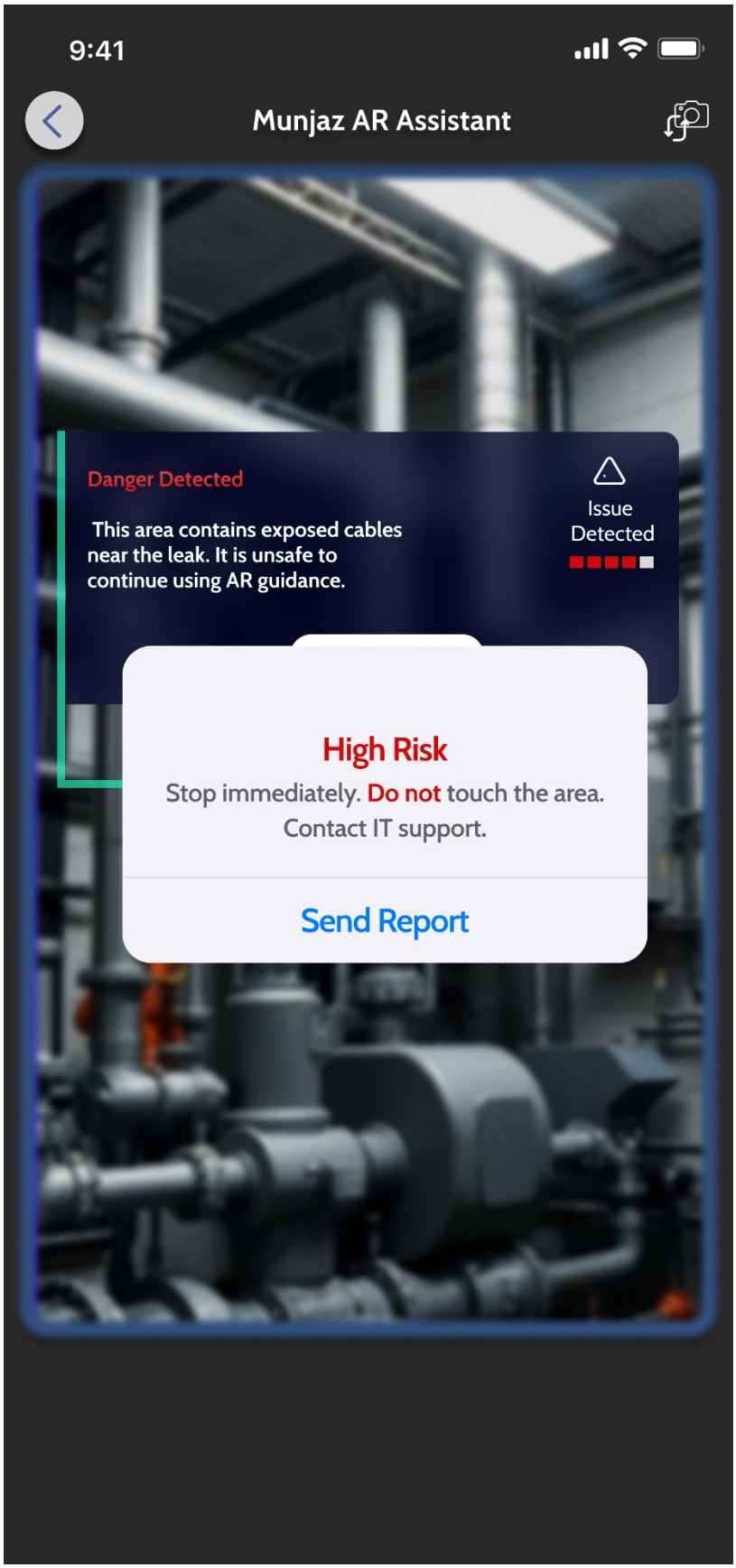
This escalation protocol effectively mitigates risk by halting the AR guidance and converting the user's device into a **data collection and reporting tool** for remote expert analysis.


9:41
Munjaz AR Assistant
Step 1 :
Move the camera closer to the floor to check the leak source.
Issue Detected
Next
Scan Again
Next


9:41
Munjaz AR Assistant
Step 2 :
Tilt the camera slightly left to inspect nearby components.
Issue Detected
Next
Scan Again
Next


9:41
Danger Detected
OK
Danger Detected
This area contains exposed cables near the leak. It is unsafe to continue using AR guidance.
Issue Detected
Next
Scan Again
Next


9:41
Danger Detected
OK
Danger Detected
This area contains exposed cables near the leak. It is unsafe to continue using AR guidance.
Issue Detected
Next
Scan Again
Next


9:41
Munjaz AR Assistant
Danger Detected
This area contains exposed cables near the leak. It is unsafe to continue using AR guidance.
Issue Detected
High Risk
Stop immediately. Do not touch the area. Contact IT support.
Send Report

Path C: Standard Resolution (Average Use Case)

This path represents the primary function of the **Munjaz AR Assistant**, guiding a user through a routine repair or maintenance task that is both **non-dangerous** and **within the scope of AR-guided fixes**. The process moves from issue identification to a step-by-step interactive fix and concludes with a successful summary report.

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1. 🛠️ Guided Repair Steps

After the initial scanning and issue identification, the system provides sequential, action-oriented instructions overlaid on the live camera view.

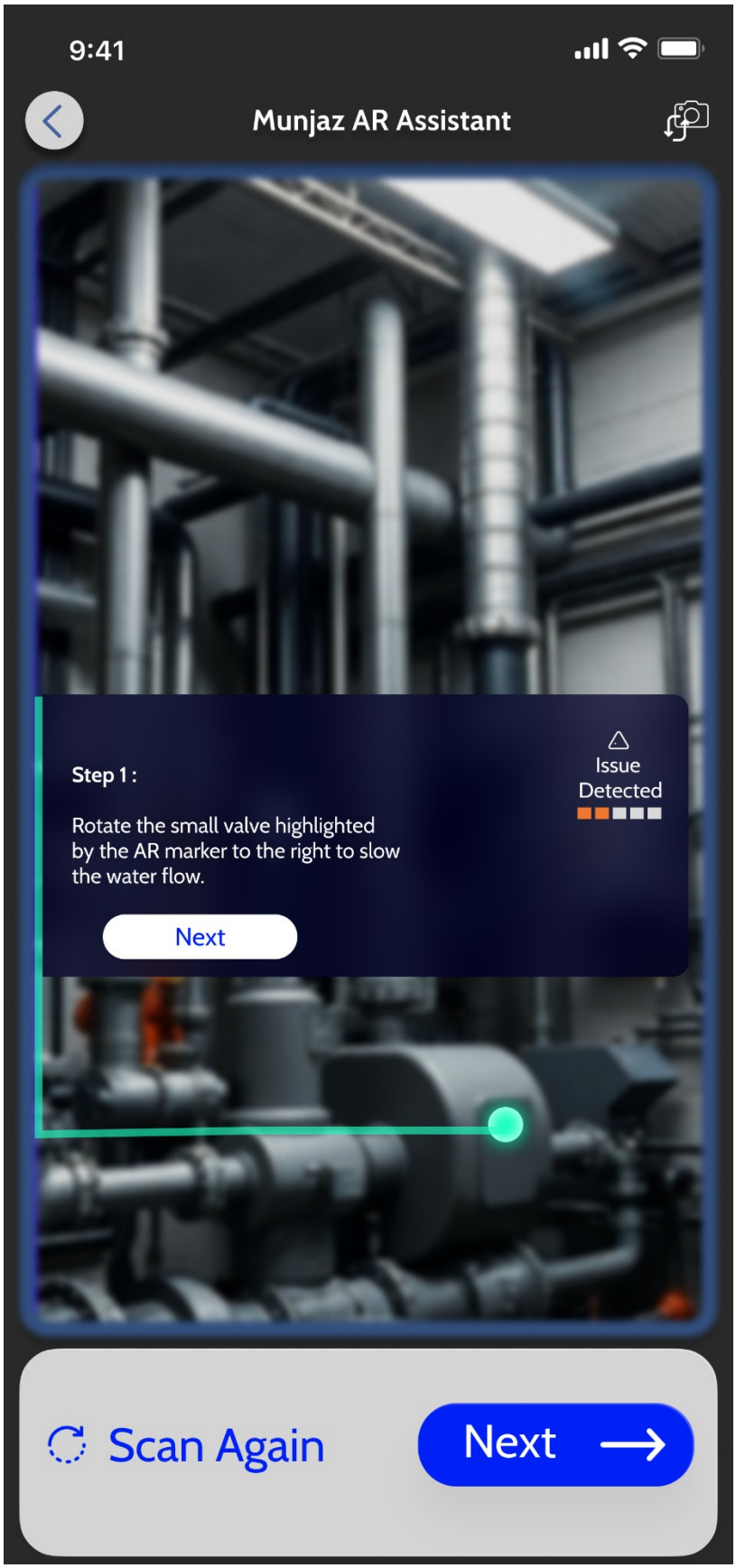
Screen Name	Description	Key Elements & Function	Transition
Munjaz AR Assistant (Step 1)	The first physical action step is presented, directing the user to manipulate a specific component.	**Guidance: Step 1: Rotate the small valve highlighted by the AR marker to the right to slow the water flow.** The AR marker visually identifies the exact valve to be turned. The **Issue Detected** indicator with its severity bar (orange/red/white) indicates that the issue is still active but not critically dangerous.	The user performs the action and taps **Next** to confirm completion and receive the next instruction.
Munjaz AR Assistant (Step 2)	The second step provides the final action required to secure the fix and stabilize the system.	**Guidance: Step 2: Press the small reset switch indicated on the screen to close the internal flow pathway.** This action finalizes the internal change needed to stop the leak. **System Feedback:** A banner appears: **The leak has stopped! and the area is now stable. No further assistance is required.** This provides instant confirmation that the user's action was successful and the environment is safe.	The user taps **Next** (or **Finish** if this is the final resolution step, as seen in) to transition to the summary report.

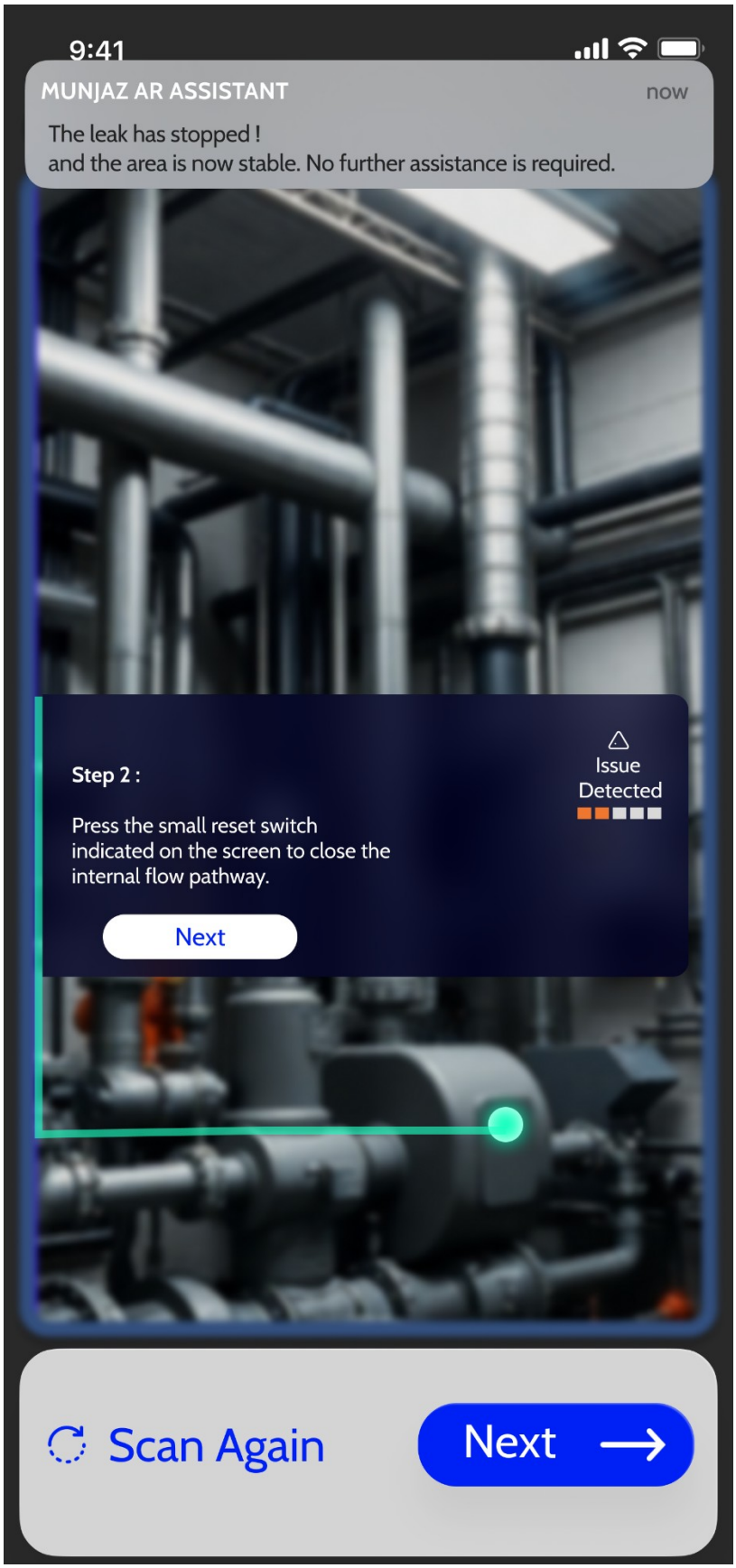
2. ✅ Session Conclusion and Documentation

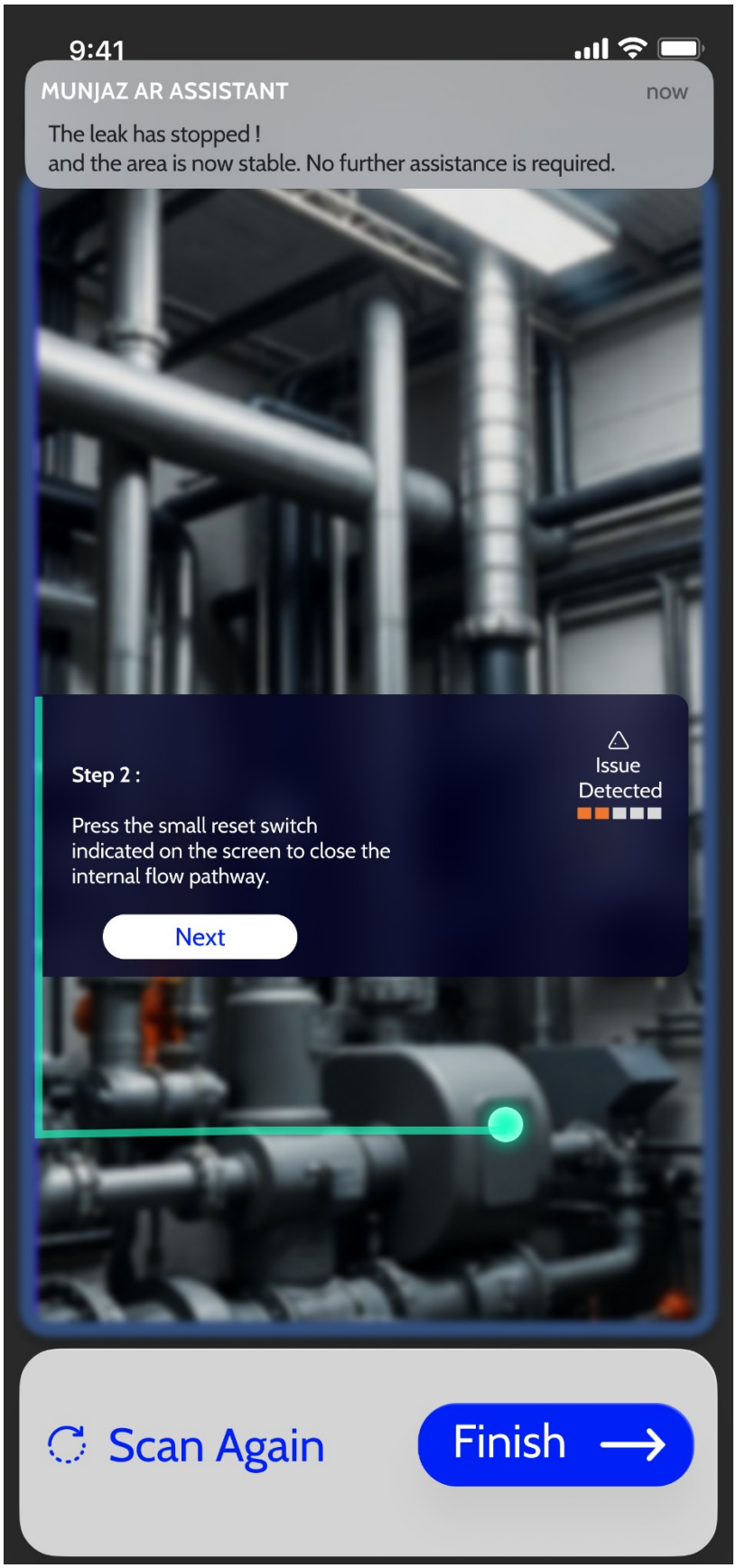
The system formally concludes the session, providing an auditable record of the repair performed.

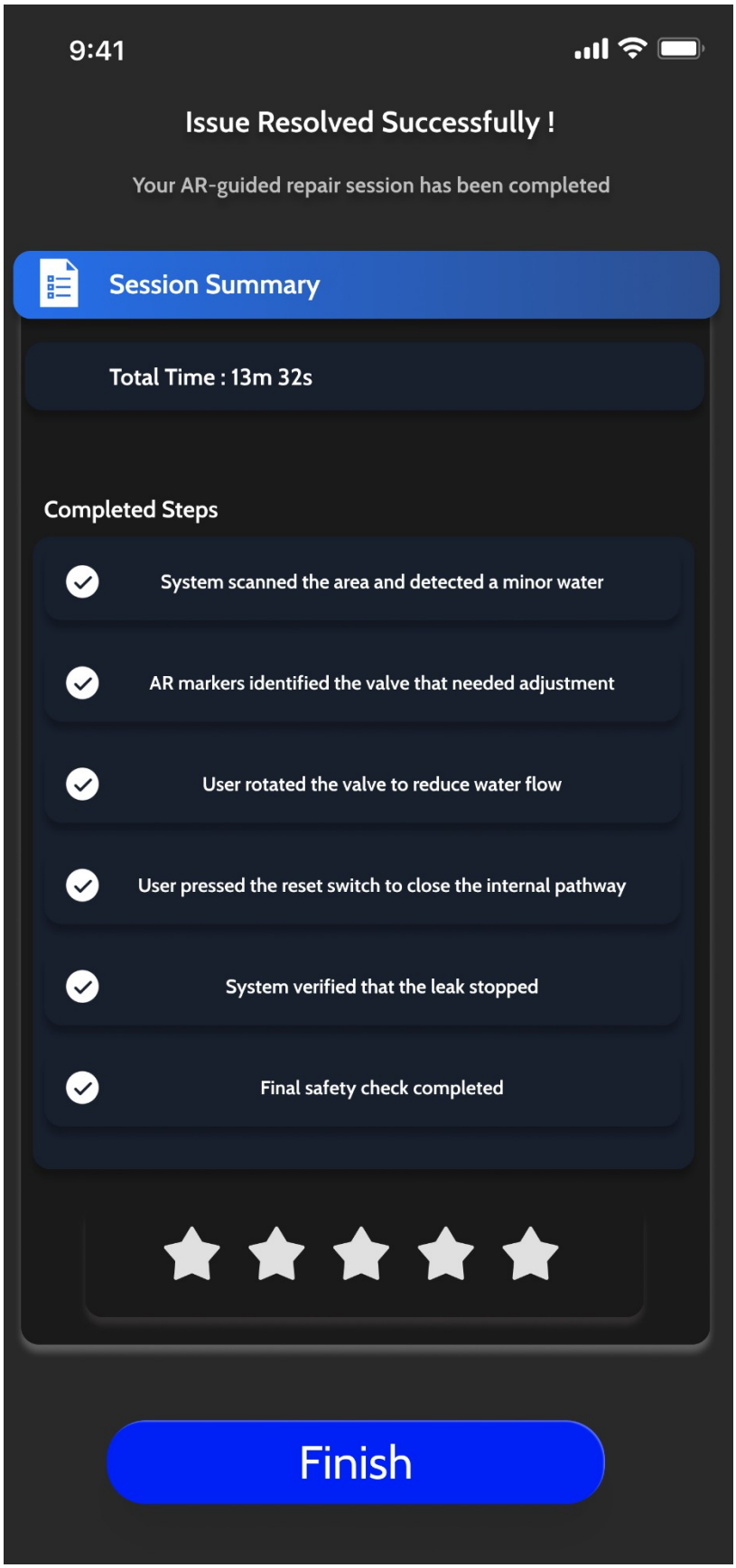
Screen Name	Description	Key Elements & Function	Outcome & Conclusion
Issue Resolved Successfully! (Session Summary)	A comprehensive report confirming the successful completion of the AR-guided repair session.	**Total Time:** Records the duration of the entire session (e.g., **13m 32s**), essential for operational metrics. **Completed Steps:** A detailed, chronological list of actions taken and verified: *System scanned*, *AR markers identified the valve*, *User rotated the valve*, *User pressed the reset switch*, *System verified that the leak stopped*, and *Final safety check completed*. **Feedback:** A 5-star rating system is included for user satisfaction feedback.	Tapping **Finish** closes the session, and the documented summary is stored for compliance and maintenance records.

This Standard Resolution Path exemplifies the efficiency and utility of AR assistance for routine maintenance, providing clear guidance, immediate confirmation, and robust documentation of the repair process.

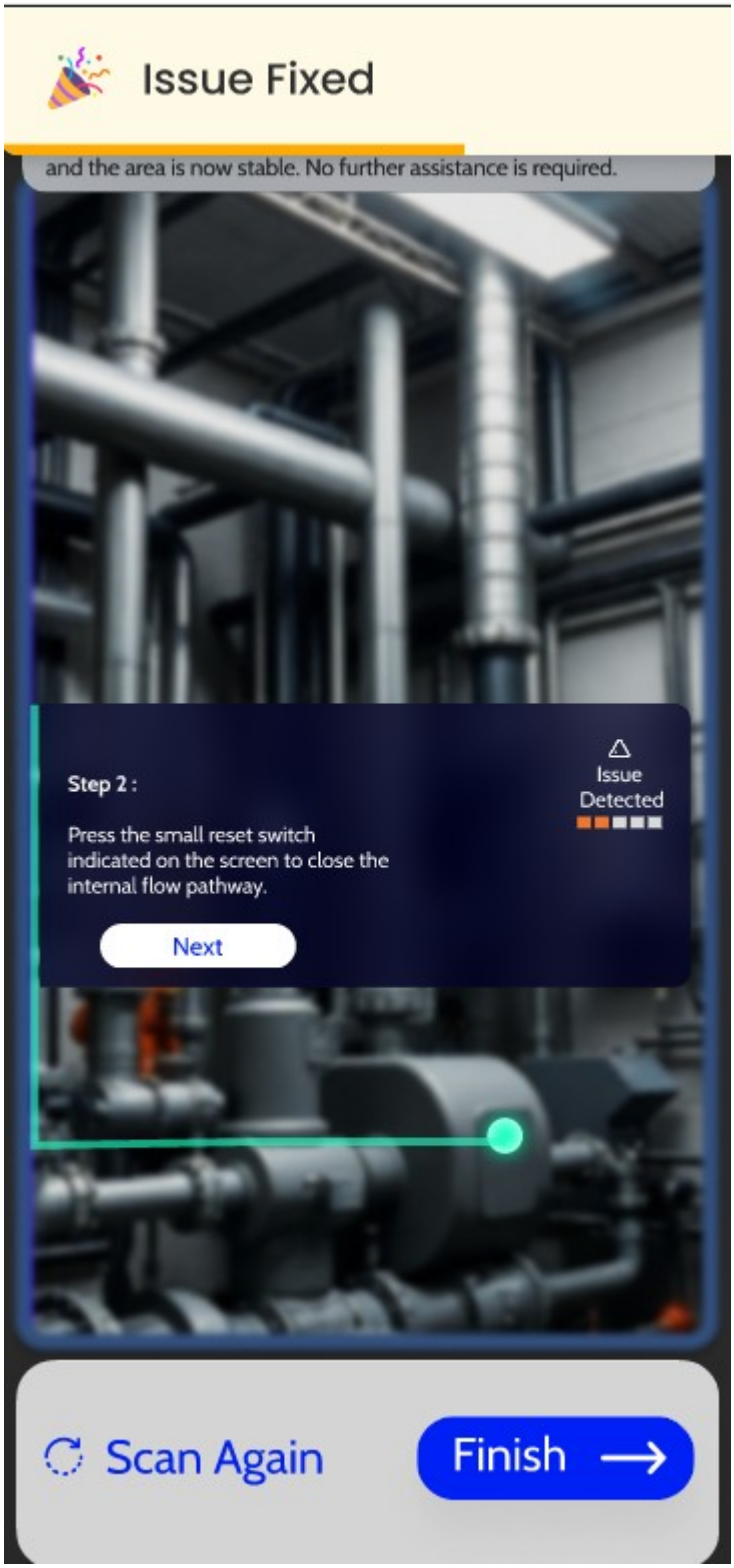

9:41
Munjaz AR Assistant
Step 1 :
Rotate the small valve highlighted by the AR marker to the right to slow the water flow.
Next
Issue Detected
Scan Again
Next


9:41
MUNJAZ AR ASSISTANT
now
The leak has stopped !
and the area is now stable. No further assistance is required.
Step 2 :
Press the small reset switch indicated on the screen to close the internal flow pathway.
Next
Issue Detected
Scan Again
Next


9:41
MUNJAZ AR ASSISTANT
now
The leak has stopped !
and the area is now stable. No further assistance is required.
Step 2 :
Press the small reset switch indicated on the screen to close the internal flow pathway.
Next
Issue Detected
Scan Again
Finish


9:41
Issue Resolved Successfully !
Your AR-guided repair session has been completed
Session Summary
Total Time : 13m 32s
Completed Steps
System scanned the area and detected a minor water
AR markers identified the valve that needed adjustment
User rotated the valve to reduce water flow
User pressed the reset switch to close the internal pathway
System verified that the leak stopped
Final safety check completed
Finish


9:41
Munjaz AR Assistant
Issue Detected
Issue Detected
Scan Again
Next


Issue Fixed
and the area is now stable. No further assistance is required.
Step 2 :
Press the small reset switch indicated on the screen to close the internal flow pathway.
Next
Issue Detected
Scan Again
Finish

General Workflow Table: Augmented Reality Guidance System

Phase	Screen Examples / Key Actions	System Functionality	Outcome & Next Phase
I. Session Initialization	**AR Guidance Screen**, **Scanning Environment**	**Readiness & Mapping:** Instructs user on operational safety and steady camera use. Performs Spatial Mapping (SLAM) to create a digital representation of the environment.	**Issue Identified:** System confirms the problem (e.g., Water Leak) and analyzes the context to determine the path (II or III).
II. Standard Resolution (Average Case)	**AR Assistant (Steps 1, 2)**, **Issue Resolved Successfully!**	**Guided Intervention:** Overlays visual markers and sequential, actionable text to guide the user (e.g., **Rotate the valve**, **Press the reset switch**). Provides real-time confirmation of the fix (e.g., **The leak has stopped!**).	**Session Completed:** Generates a **Session Summary** with total time and documented steps for compliance and record-keeping.
III. Safety Escalation (Dangerous Case)	**AR Assistant (Step 1/2)**, **Danger Detected**, **High Risk**	**Hazard Detection:** Utilizes computer vision to detect co-located environmental risks (e.g., exposed cables near water). **Safety Override:** Immediately stops repair guidance and displays a critical **Red Banner** warning.	**Expert Hand-off:** Mandates immediate cessation (**Do not touch**) and provides an interface to **Send Report** to IT support for remote escalation and expert intervention.